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Problem 1. *Consider the following probit model:*

$$d = \mathbf{1}\{\theta_0' x \geq \varepsilon\}, \qquad \varepsilon | x \sim N(0,1), \qquad \theta_0 \in \Theta \subset \mathbb{R}^p.$$

x is an $\mathbb{R}^p$-valued random variable, drawn from an unspecified distribution. The econometrician observes a random sample $\{d_i, x_i\}$, $i = 1, \ldots, n$ generated according to the above model, and wishes to estimate θ_0 using GMM. In particular, let

$$g(z, \theta) = a(x)(d - \Phi(x'\theta))$$

where Φ denotes the standard normal CDF and $a(x)$ is a q-dimensional vector of functions of x, then define $\bar{g}(\theta) = \frac{1}{n}\sum_{i=1}^n g(z_i, \theta)$. The estimator $\hat{\theta}$ minimizes $Q_n(\theta) = \bar{g}(\theta)'\bar{g}(\theta)$ (that is, the weighting matrix is an identity matrix). Assume that Θ is compact. Also, let $a_j(x)$ denote the j-th element of $a(x)$ and assume that $E[|a_j(x)|] < \infty$, $j = 1, \ldots, q$. Finally, assume that $Q(\theta) = E[g(z,\theta)]'E[g(z,\theta)]$ has a unique minimizer. Discuss the consistency property of $\hat{\theta}$.

Solution. The consistency of the GMM estimator $\hat{\theta}$ which minimizes $Q_n(\theta)$ is guaranteed by the following conditions for extremum estimators: (i) Compactness: The parameter space Θ is compact by assumption. (ii) Identification: The population objective function $Q(\theta)$ is assumed to have a unique minimizer θ_0 in Θ, where $E[g(z, \theta_0)] = 0$ by construction of the GMM moment condition. (iii) Uniform Convergence: The sample objective function $Q_n(\theta)$ converges uniformly in probability to the population objective function $Q(\theta)$ over Θ, i.e., $\sup_{\theta \in \Theta} |Q_n(\theta) - Q(\theta)| \xrightarrow{p} 0$. The moment condition $E[|a_j(x)|] < \infty$ implies $E[\sup_{\theta \in \Theta} |g_j(z, \theta)|] < \infty$ since $|d - \Phi(x'\theta)| \leq 1$ is bounded, ensuring the Uniform Law of Large Numbers applies to justify uniform convergence. Given these conditions, the GMM estimator is consistent, $\hat{\theta} \xrightarrow{p} \theta_0$. □

Problem 2. *Suppose the conditional density of z given x takes the form $f(z|x, \theta_0)$, $\theta_0 \in \Theta \subset \mathbb{R}^k$, where $f(z|x, \theta_0)$ is a parametric density function. This can be used to calculate the conditional MLE for θ_0.*

On the other hand, suppose we can find an appropriate function g such that

$$E[g(z, \theta_0)|x] = 0, \quad \theta_0 \in \Theta$$

where z and x are observable random variables, and g is a function of z and is known up to the finite dimensional unknown parameter θ_0. This conditional moment restriction provides alternative methods (MOM, GMM) to estimate θ_0. Note that any function of x, $a(x)$, say, serves as an instrumental variable, since

$$E[a(x)g(z, \theta_0)] = 0$$

by the law of iterated expectation. It is known that the optimal instrument is

$$a^*(x) = D(x)'V(x)^{-1}$$

where $V(x) = E[g(z,\theta_0)g(z,\theta_0)'|x]$ and $D(x) = E[\frac{\partial g(z,\theta_0)}{\partial \theta}|x]$. (Moreover, the resulting estimator achieves the "semiparametric efficiency bound," as shown by Chamberlain, 1987, JOE. Kitamura, Tripathi and Ahn (Ecta 2004) provide a practical empirical likelihood-based procedure that achieves this bound.) Show that the conditional MLE is asymptotically no less efficient than the optimal IV estimator (whose asymptotic variance is $[E[D'(x)V(x)^{-1}D(x)]]^{-1}$).

Solution. The asymptotic variance of the conditional MLE is $V_{C-MLE} = I(\theta_0)^{-1}$, where $I(\theta_0) = E[-\frac{\partial^2 \log f(z|x,\theta_0)}{\partial\theta\partial\theta'}]$ is the Fisher information matrix. The asymptotic variance of the optimal IV estimator is $V^*_{IV} = [E[D(x)'V(x)^{-1}D(x)]]^{-1}$.

We show the result using the generalized information equality for conditional moment restriction models. For the conditional moment restriction $E[g(z,\theta_0)|x] = 0$, the moment-based estimator using optimal instruments $a^*(x) = D(x)'V(x)^{-1}$ has asymptotic variance $[E[D(x)'V(x)^{-1}D(x)]]^{-1}$. The conditional score from the full likelihood is $s(z,\theta_0) = \frac{\partial \log f(z|x,\theta_0)}{\partial\theta}$ with $E[s(z,\theta_0)|x] = 0$. Since any valid moment function $g(z,\theta)$ with $E[g(z,\theta_0)|x] = 0$ must lie in the tangent space spanned by the conditional score, we can write $g(z,\theta_0) = A(x)s(z,\theta_0) + \eta(z,x)$ where $E[\eta(z,x)|x] = 0$ and η is orthogonal to s conditional on x.

Taking $D(x) = E[\frac{\partial g(z,\theta_0)}{\partial\theta}|x] = A(x)E[\frac{\partial s(z,\theta_0)}{\partial\theta}|x] = -A(x)I(x)$ and

$$V(x) = E[g(z,\theta_0)g(z,\theta_0)'|x] = A(x)I(x)A(x)'$$

where $I(x) = E[s(z,\theta_0)s(z,\theta_0)'|x]$ is the conditional information. The information from the moment restriction is

$$E[D(x)'V(x)^{-1}D(x)] = E[I(x)A(x)'[A(x)I(x)A(x)']^{-1}A(x)I(x)].$$

By the Cauchy-Schwarz inequality for matrices, this is bounded above by $E[I(x)] = I(\theta_0)$, with equality only when $A(x)$ projects onto the full score space, i.e., when the moment restriction captures all information in the likelihood. Therefore $I(\theta_0) \geq E[D(x)'V(x)^{-1}D(x)]$, and taking inverses, $V_{C-MLE} = I(\theta_0)^{-1} \leq [E[D(x)'V(x)^{-1}D(x)]]^{-1} = V^*_{IV}$. The conditional MLE is asymptotically no less efficient. □

Problem 3. *Consider a censored regression model*

$$y = \max\{\theta_{01}'x + \varepsilon,\ 0\}, \qquad \theta_{01} \in \mathbb{R}^k$$

with $\varepsilon|x \sim N(0,\sigma^2)$. Let d_i be an indicator taking the value of 1 when the i-th observation is uncensored and 0 otherwise. We have seen that $E[y|x,d=1] = \theta_{01}'x + \theta_{02}\lambda(\gamma_0'x)$ where $\lambda(\cdot)$ is the inverse Mills ratio ϕ/Φ and $\gamma_0 = \theta_{01}/\theta_{02}$ (neither the specific functional form of λ nor the fact that γ_0 is a function of θ's is important in answering this question), which motivates the well-known two step procedure to obtain $(\hat\theta_1, \hat\theta_2)$ as the minimizer of

$$\frac{1}{n}\sum_{i=1}^n d_i[y_i - \theta_1'x_i - \theta_2\lambda(\hat\gamma'x_i)]^2$$

where the Probit MLE $\hat\gamma$ maximizes $\sum_{i=1}^n [d_i \log\Phi(\gamma'x_i) + (1-d_i)\log(1-\Phi(\gamma'x_i))]$.

As we discussed in class, the asymptotic variance of $\hat\theta$ takes the form

$$E\Big[\frac{\partial^2}{\partial\theta\partial\theta'}g(z,\theta_0,\gamma_0)\Big]^{-1}\Omega E\Big[\frac{\partial^2}{\partial\theta\partial\theta'}g(z,\theta_0,\gamma_0)\Big]^{-1}$$

where Ω is given by

$$E\Big[\Big(\frac{\partial}{\partial\theta}g(z,\theta_0,\gamma_0) + E\Big[\frac{\partial^2}{\partial\theta\partial\gamma'}g(z,\theta_0,\gamma_0)\Big]\psi(z)\Big)\Big(\frac{\partial}{\partial\theta}g(z,\theta_0,\gamma_0) + E\Big[\frac{\partial^2}{\partial\theta\partial\gamma'}g(z,\theta_0,\gamma_0)\Big]\psi(z)\Big)'\Big].$$

Obtain the matrices $E[\frac{\partial^2}{\partial\theta\partial\theta'}g(z,\theta_0,\gamma_0)]$, $E[\frac{\partial^2}{\partial\theta\partial\gamma'}g(z,\theta_0,\gamma_0)]$ as well as the functions $\frac{\partial}{\partial\theta}g(\cdot,\theta_0,\gamma_0)$ and $\psi(\cdot)$ for the above two step estimator. (You can write your answer in terms of λ; no need to write it in terms of ϕ and Φ.)

Solution. Let $r(y,x;\theta,\gamma) = y - \theta_1'x - \theta_2\lambda(\gamma'x)$, $X_\lambda(x,\gamma) = \begin{pmatrix} x \\ \lambda(\gamma'x) \end{pmatrix}$, $u = \gamma'x$, and $K(u) = \lambda(u)(\lambda(u)-u)$ so $\lambda'(u) = -K(u)$. Take the moment function $g(z,\theta,\gamma) = -d\, r(y,x;\theta,\gamma)\, X_\lambda(x,\gamma)$.

$\frac{\partial g}{\partial\theta'}(z,\theta_0,\gamma_0) = d\, X_\lambda(x,\gamma_0)\, X_\lambda(x,\gamma_0)'$. Hence

$$E\left[\frac{\partial^2}{\partial\theta\partial\theta'}g(z,\theta_0,\gamma_0)\right] = E\left[d\begin{pmatrix} xx' & x\lambda(\gamma_0'x) \\ \lambda(\gamma_0'x)x' & \lambda^2(\gamma_0'x)\end{pmatrix}\right].$$

$\frac{\partial g}{\partial\gamma'}(z,\theta,\gamma) = -d\left[\frac{\partial r}{\partial\gamma'}X_\lambda + r\frac{\partial X_\lambda}{\partial\gamma'}\right]$, with $\frac{\partial r}{\partial\gamma'} = -\theta_2\lambda'(\gamma'x)x' = \theta_2 K(\gamma'x)x'$ and

$$\frac{\partial X_\lambda}{\partial\gamma'} = \begin{pmatrix} 0 \\ \lambda'(\gamma'x)x' \end{pmatrix} = \begin{pmatrix} 0 \\ -K(\gamma'x)x' \end{pmatrix}.$$

Evaluating at (θ_0,γ_0) and using $E[d\, r \mid x] = 0$ gives

$$E\left[\frac{\partial^2}{\partial\theta\partial\gamma'}g(z,\theta_0,\gamma_0)\right] = -\theta_{02}\, E\left[\, d\, K(\gamma_0'x)\, X_\lambda(x,\gamma_0)\, x'\right].$$

The per-observation derivative appearing in Ω is

$$\frac{\partial}{\partial\theta}g(z,\theta_0,\gamma_0) = -d\, r(\theta_0,\gamma_0)\, X_\lambda(x,\gamma_0).$$

The first-step influence function (Probit MLE): the individual score is

$$s(z) = \frac{\partial\ell_i^p(\gamma_0)}{\partial\gamma} = x\,\frac{\phi(\gamma_0'x)\,[d - \Phi(\gamma_0'x)]}{\Phi(\gamma_0'x)\,[1-\Phi(\gamma_0'x)]},$$

the Fisher information is

$$I_p = E\left[-\frac{\partial^2\ell_i^p(\gamma_0)}{\partial\gamma\partial\gamma'}\right] = E\left[\frac{\phi(\gamma_0'x)^2}{\Phi(\gamma_0'x)\,[1-\Phi(\gamma_0'x)]}\, xx'\right], \qquad \psi(z) = I_p^{-1}s(z).$$

□

Problem 4. *(Poisson Regression) We observe (y_i,x_i), $i = 1,\dots,n$, $y_i \in \mathbb{N}$, $x_i \in \mathbb{R}^k$, that are IID random variables. The conditional density of y given x is*

$$f(y_i|x_i) = \frac{\exp(y_i x_i'\beta_0 - \exp(x_i'\beta_0))}{y_i!}.$$

The density of x_i is left unspecified. Assume that $E[\exp(x_i'\beta)] < \infty$ for all $\beta \in \mathbb{R}^k$ and $E[x_i x_i']$ is (finite and) nonsingular. It is easy to show that

$$E[y_i|x_i] = \exp(x_i'\beta_0), \qquad E[y_i^2|x_i] = \exp(x_i'\beta_0) + \exp(2x_i'\beta_0);$$

you can use these results to answer the following questions.

(i) Is the (conditional) ML for $\beta_0 \in \mathbb{R}^k$ consistent? If so, provide a proof (including the identifiability proof). Note that the parameter space is not assumed to be compact.

(ii) Now assume that the parameter space (B, say) for β is compact. Consider the NLLS estimator that minimizes $Q_n(\beta) = \frac{1}{n}\sum_{i=1}^n (y_i - \exp(x_i'\beta))^2$ over B. Show the consistency of NLLS (if you need additional assumptions to prove it, state them explicitly in your answer).

(iii) Derive the asymptotic distribution of the NLLS estimator.

Solution.

1. The Conditional MLE $\hat{\beta}$ is consistent. *Identifiability:* The population log-likelihood $Q(\beta) = E[l_i(\beta)]$ has the Hessian $\nabla^2 Q(\beta) = -E[\exp(x_i'\beta)x_i x_i']$. Since $\exp(x_i'\beta) > 0$ and $E[x_i x_i']$ is nonsingular, $\nabla^2 Q(\beta)$ is negative definite. This strict concavity of $Q(\beta)$ ensures β_0 is the unique global maximizer, satisfying the identification condition. Given this identification and standard regularity conditions for MLE consistency (continuity, dominated convergence, integrability from the moment assumptions), the MLE $\hat{\beta}$ is consistent by the standard extremum estimator theorem for MLEs.

2. The NLLS estimator $\hat{\beta}_{NLLS}$ is consistent. *Additional assumption:* $E[y_i^2] < \infty$, which follows from the given conditional moments. The identification condition is satisfied because by the law of iterated expectations, the population objective $Q(\beta) = E[(y_i - \exp(x_i'\beta))^2]$ is uniquely minimized at β_0 where $\exp(x_i'\beta_0) = E[y_i|x_i]$ almost surely. Given nonsingularity of $E[x_i x_i']$, this implies $\beta = \beta_0$ uniquely. The NLLS consistency follows from the extremum estimator theorem: β_0 is the unique minimizer of $Q(\beta)$ on the compact set B, and uniform convergence $Q_n(\beta) \xrightarrow{p} Q(\beta)$ holds by the Uniform Law of Large Numbers applied to the continuous objective with finite moments.

3. The asymptotic distribution of the NLLS estimator is $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(\mathbf{0}, V)$, where $V = \Gamma^{-1}\Omega\Gamma^{-1}$. The first-order condition is $\frac{1}{n}\sum_{i=1}^n (y_i - \exp(x_i'\beta))\exp(x_i'\beta)x_i = 0$. Define the moment function $m_i(\beta) = (y_i - \exp(x_i'\beta))\exp(x_i'\beta)x_i$. The Jacobian is $\frac{\partial m_i(\beta)}{\partial \beta'} = [(y_i - \exp(x_i'\beta)) - \exp(x_i'\beta)]\exp(x_i'\beta)x_i x_i'$. At β_0, taking expectations and using $E[y_i|x_i] = \exp(x_i'\beta_0)$ gives $\Gamma = -E\left[\frac{\partial m_i(\beta_0)}{\partial \beta'}\right] = E[\exp(2x_i'\beta_0)x_i x_i']$. For the variance, $\Omega = E[m_i(\beta_0)m_i(\beta_0)'] = E[(y_i - \exp(x_i'\beta_0))^2\exp(2x_i'\beta_0)x_i x_i']$. Using $E[(y_i - \exp(x_i'\beta_0))^2|x_i] = \mathrm{Var}(y_i|x_i) = \exp(x_i'\beta_0)$ gives $\Omega = E[\exp(3x_i'\beta_0)x_i x_i']$. Therefore,

$$V = \left(E[\exp(2x_i'\beta_0)x_i x_i']\right)^{-1} E[\exp(3x_i'\beta_0)x_i x_i'] \left(E[\exp(2x_i'\beta_0)x_i x_i']\right)^{-1}.$$

□

Problem 5. *A data set from the 1963 Survey of Consumer Finances (SCF), adapted from Mirer (1988) is posted on the class web page. There are 11 series (plus ID numbers) for 100 samples (family heads) in the data set. They are: V1: ID number, V2: family size, V3: education (years), V4: age (years), V5: experience (years), V6: months worked, V7: race (1 = white, 2 = black), V8: region (1 = northeast, 2 = northcentral, 3 = south, 4 = west), V9: earnings ($1000), V10: income ($1000), V11: wealth ($1000), V12: savings ($1000)*

Using the SCF data set, create an "artificial" binary data series y_i, $i = 1, \ldots, 100$ such that $y_i = 1$ if earnings > 9 (thousand dollars), $y_i = 0$ otherwise (do not do such a thing in practice). Use a constant, Education, Experience, Region as covariates (see the first section of the attached program). We estimate the logit model $P\{y_i = 1|x_i\} = F(\beta_0' x_i) = \frac{e^{\beta_0' x_i}}{1+e^{\beta_0' x_i}}$, where $x_i' = (1, Education, Experience, Region)_i$ is the vector of characteristics of Household i and β_0 is the vector of the true parameters. Assume that the data are IID. Attached is a sample code in Python (you can use other languages as well, though in that case make sure to write your own Newton iteration code just like the sample code does).

(i) Let $l(\beta) = \sum_{i=1}^n l_i(\beta)$ be the log likelihood function at β, where $l_i(\beta)$ denotes the contribution of the i-th observation to $l(\beta)$. Show that the score $\partial l_i(\beta)/\partial\beta$ of the Logit model satisfies the relation: $\frac{\partial l_i(\beta)}{\partial \beta} = (y_i - F(\beta' x_i))x_i$. Conclude that the (observed) negative Hessian is $-\frac{\partial^2 l(\beta)}{\partial\beta\partial\beta'} = \sum_{i=1}^n f(\beta' x_i)x_i x_i'$, where $f(\cdot)$ denotes the density function of the standard logistic distribution.

(ii) Calculate the log-likelihood at $\beta = (0, 0, 0, 0, 0, 0)'$.

(iii) Obtain the MLE for β and the standard error for each MLE estimate, using the "outer-product (of the gradient)" form for maximization and covariance estimation: see the sample code.

(iv) Modify your code so that the formula (1) you derived above is used for optimization and the covariance estimation. Report the new MLEs (should be close to the former estimate) and their standard errors (should be somewhat different from the former standard errors).

Solution. (i) The score for observation i is:

$$\frac{\partial l_i(\beta)}{\partial \beta} = y_i \frac{1}{F(\beta' x_i)} \frac{\partial F(\beta' x_i)}{\partial \beta} - (1-y_i) \frac{1}{1 - F(\beta' x_i)} \frac{\partial F(\beta' x_i)}{\partial \beta} = \left[\frac{y_i(1-F) - (1-y_i)F}{F(1-F)} \right] f(\beta' x_i) x_i$$

Since $f(\beta' x_i) = F(\beta' x_i)(1 - F(\beta' x_i))$, the expression simplifies to: $\frac{\partial l_i(\beta)}{\partial \beta} = [y_i - F(\beta' x_i)]x_i$.

The negative Hessian of the total log-likelihood is derived from the second derivative:

$$-\frac{\partial^2 l(\beta)}{\partial\beta\partial\beta'} = -\sum_{i=1}^n \frac{\partial^2 l_i(\beta)}{\partial\beta\partial\beta'} = -\sum_{i=1}^n [-f(\beta' x_i)x_i x_i'] = \sum_{i=1}^n f(\beta' x_i)x_i x_i'.$$

(ii) At $\beta = \mathbf{0}$, $F(\beta' x_i) = 0.5$.

$$l(\mathbf{0}) = 100 \times \ln(0.5) \approx -69.3147$$

(iii) & (iv) The MLE coefficients ($\hat{\beta}$) and their standard errors (SE) are shown below in the format: Estimate/(Standard Error). Both methods converge to the same final log-likelihood $\ln L(\hat{\beta}) = -36.3819$.

```
import numpy as np
import pandas as pd
data = pd.read_csv('metdata.txt', sep=r'\s+')
n = len(data)
y = (data['V9'] > 9).astype(int).values
x = np.column_stack([np.ones(n), data['V3'], data['V5']] + \
                    [np.where(data['V8'] == i, 1, 0) for i in [2,3,4]])
def ll(p, y):
    return np.sum(y * np.log(np.clip(p, 1e-15, 1-1e-15)) + \
                  (1-y) * np.log(1 - np.clip(p, 1e-15, 1-1e-15)))
p_res = 1 / (1 + np.exp(-x @ np.zeros(6)))
lk_res = ll(p_res, y)
def mle(x, y, method='hessian'):
    c = np.zeros(6)
    for _ in range(20):
        p = 1 / (1 + np.exp(-x @ c))
        g = (y - p).reshape(-1,1) * x
        xu = g.sum(0)
        if method == 'opg':
            I = g.T @ g
        else:
            I = x.T @ ((p*(1-p)).reshape(-1,1) * x)
        try:
            dc = np.linalg.inv(I) @ xu
            c += dc
        except:
            break
        if np.abs(dc).max() < 1e-5:
            break
    se = np.sqrt(np.diag(np.linalg.inv(I)))
    return c, se, ll(p, y)
opg_c, opg_se, lk_opg = mle(x, y, 'opg')
hess_c, hess_se, lk_hess = mle(x, y, 'hessian')
```

```
Variable         OPG Estimate (SE)  Hessian Estimate (SE)
------------------------------------------------------------
Constant             -8.2509 (2.3433)  -8.2512 (2.2537)
Education (V3)        0.5387 (0.1326)   0.5387 (0.1267)
Experience (V5)       0.0378 (0.0470)   0.0378 (0.0413)
Region 2 (NC)        -0.3987 (0.7812)  -0.3986 (0.7533)
Region 3 (South)     -1.2469 (0.9362)  -1.2470 (0.8569)
Region 4 (West)      -1.1415 (0.8906)  -1.1413 (0.8855)
```

□